AI-Powered E-Commerce Scraper

Replacing XPath/CSS Selector Maintenance with LLM Extraction (Ollama)

Problem Statement

- Traditional scrapers string extractors depend on XPath/CSS selectors
- E-commerce sites frequently change DOM structure
- Small UI changes break extraction logic
- Fixing selectors becomes repeated maintenance work

Developer Pain Point

- Time wasted debugging broken selectors
- Same site requires continuous updates
- Frequent re-deployments and re-testing
- Less time for real product development

Business Impact

- High maintenance cost
  (developers time waste, waste of resources using for deployments and testing again and again for the same site)
- Delayed updates for customers

  (gives frustration for customers due to frequent breaking of processors)

Proposed Solution

- Scrape HTML normally (Playwright)
- Clean + reduce HTML (Cheerio)
- Send cleaned HTML to AI model
- AI returns structured product JSON

Architecture (High Level)

- API Request (URL)
- Playwright Scraper → HTML
- HTML Cleaner → Reduced HTML
- Ollama LLM Extractor → JSON
- Zod Validation + Normalization
- Response + Logging

Demo Website Used

- books.toscrape.com (stable demo site)
- Extracted fields: title, price, rating, stock, description, image, UPC
- Output: Strict validated JSON response

Cost Effectiveness

- Ollama runs locally (no paid API usage)
- Cost is only compute + infrastructure
- Avoids repeated developer maintenance cost
- Scales better than manual selector updates

Future Scope: Train Our Own Model

- We already collect large HTML datasets
- Store HTML + correct extracted product JSON
- Use this as training data for fine-tuning
- Build a dedicated HTML-to-Product extraction model

Conclusion

- Selectors are brittle and costly to maintain
- AI extraction reduces repetitive maintenance
- Validation ensures reliable structured output
- Developers can focus on solving real problems